

KNOWLEDGE ENGINEERING AND ONTOLOGIES COURSE PROJECT REPORT

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1. Executive Summary

This project presents the development of a semantic ontology for an online bookstore using Semantic Web technologies. The main objective is to organize and represent bookstore knowledge in a structured way to support intelligent querying, semantic search, and recommendation systems. The ontology was developed using Protégé and implemented with OWL and RDF standards. It models concepts such as books, authors, customers, categories, orders, reviews, publishers, and payments. Data was collected from publicly available bookstore information and manually created educational examples. The project also includes knowledge graph construction, SPARQL querying, and ontology validation principles. In addition, recent research about ontology population using Large Language Models (LLMs) was studied and integrated conceptually into the project. This ontology can support future intelligent bookstore applications by improving information organization and enabling semantic reasoning.

2. Description of the Project

The objective of this project is to build a semantic ontology for an online bookstore system. The ontology aims to organize information related to books, authors, customers, categories, reviews, publishers, orders, and payments.

The project addresses the problem of unstructured information management in online bookstores. Traditional databases store information efficiently but do not provide semantic understanding between entities. Ontologies solve this problem by explicitly representing concepts and relationships.

The project uses ontology engineering principles, Semantic Web technologies, OWL modeling, RDF representation, SPARQL querying, and knowledge graph construction.

The project was implemented using:

- Protégé
- GraphDB
- OWL
- RDF
- SPARQL
- GitHub
- Widoco

The target users are:

- student
- bookstore administrators
- researchers
- semantic web engineers

The expected outputs are:

- OWL ontology
- RDF knowledge graph
- SPARQL queries
- SHACL validation rules
- Widoco documentation
- GitHub repository

Competency Questions

1. Which books belong to the Horror category?
2. Which author wrote Harry Potter?
3. Which customer added books to favorites?
4. What books are currently in the shopping cart?
5. Which books were purchased by a customer?
6. What is the price of The Silent Patient?
7. Which category contains the most books?
8. Which customer completed a payment?

3. Ontology Design

TBox (Conceptual Schema)

The TBox defines the conceptual structure of the ontology.

Classes

- Book
- Author
- Category
- Customer
- Cart
- Order
- Payment
- Favorite
- Review
- Publisher
- Wishlist
- Recommendation

Object Properties

- writtenBy
- belongsToCategory

- addedToFavorites
- addedToCart
- hasOrder
- hasPayment
- hasReview
- publishedBy
- hasWishlist
- recommendedTo

Data Properties

- bookTitle
- bookPrice
- authorName
- customerName
- paymentMethod
- orderDate
- reviewText
- rating
- publisherName

Restrictions

Book:

- writtenBy some Author
- belongsToCategory some Category
- publishedBy some Publisher

Customer:

- hasOrder some Order
- addedToCart some Book
- hasWishlist some Wishlist

Order:

- hasPayment some Payment

Review:

- reviewOf some Book

ABox (Instances)

Books

- HarryPotter
- AtomicHabits

- PrideAndPrejudice
- SilentPatient
- TheAlchemist

Authors

- JkRowling
- JamesClear
- JaneAusten
- AlexMichaelides
- PauloCoelho

Categories

- Fantasy
- SelfDevelopment
- Romance
- Thriller
- Adventure

Customers

- Zaineb
- Fatma
- Ahmed

Publishers

- PenguinBooks
- HarperCollins

Reviews

- Review1
- Review2
- Review3
- Review4
- Review5

4. Data Acquisition

The data used in this project was obtained through manually created sample data and publicly available online bookstore information.

The collected data includes:

- books
- authors
- categories

- customers
- orders
- payments
- reviews
- publishers
- wishlists

The information was initially organized in table structures and transformed into ontology individuals.

Data formats include:

- CSV-like structures
- text descriptions
- OWL
- RDF

Data preprocessing

The following preprocessing steps were applied:

- removing duplicate entries
- correcting inconsistent naming
- standardizing category names
- organizing entities into ontology classes
- creating semantic relationships
- transforming data into RDF triples

Data quality and limitations

The dataset is educational and manually created. It is relatively small and does not represent all real-world bookstore complexities.

5. Knowledge Graph Constuction:

The ontology was imported into GraphDB to build a knowledge graph.

Steps performed in GraphDB:

1. Created a new repository
2. Imported OWL/RDF file
3. Loaded triples into graph
4. Executed SPARQL queries

Example RDF triples:

- HarryPotter writtenBy JkRowling
- HarryPotter belongsToCategory Fantasy
- HarryPotter hasReview Review1
- AtomicHabits writtenBy JamesClear

6. SHACL Validation:

SHACL was used to validate ontology consistency.

Example SHACL rules:

- Every Book must have at least one author
- Every Book must belong to a category
- Rating must be between 1 and 5
- Customer must have valid orders
- **Example SHACL shape:**
- ```
ex:BookShape
 a sh:NodeShape ;
 sh:targetClass ex:Book ;
 sh:property [
 sh:path ex:writtenBy ;
 sh:minCount 1 ;
] ;
 sh:property [
 sh:path ex:belongsToCategory ;
 sh:minCount 1 ;
] .
```

## 7. SPARQL Queries:

### Query 1: List all books

```
SELECT ?book WHERE {
 ?book a :Book .
}
```

### Query 2: Find authors of books

```
SELECT ?book ?author WHERE {
 ?book :writtenBy ?author .
}
```

### Query 3: Books by category

```
SELECT ?book ?category WHERE {
 ?book :belongsToCategory ?category .
}
```

### Query 4: Books in cart

```
SELECT ?customer ?book WHERE {
 ?customer :addedToCart ?book .
}
```

## 8. Widoco Documentation:

The ontology documentation was generated using WIDOCO.

It includes:

- Ontology metadata
- Class hierarchy
- Object properties
- Data properties
- RDF visualization

- Usage examples  
<https://github.com/your-repo/widoco/index.html>

## **9. Research Integration**

### **Selected Research**

Ontology Population using Large Language Models (LLMs)

#### **Why this study was selected**

This study was selected because it demonstrates how AI can automatically populate ontologies.

#### **Relevance**

The study is relevant because it can automatically classify books, authors, and categories.

#### **Integration**

The research can be integrated into:

- semantic recommendation systems
- ontology population
- semantic enrichment
- intelligent querying

## **10. Mandatory Paper Review**

Paper:

Personalized Ontology and Deep Training Tree-Based Optimal GRU-RNN for Prediction of Students' Behavior

### **Summary**

The paper combines ontology and deep learning to predict user behavior.

### **Key Contributions**

- semantic modeling
- ontology integration
- intelligent prediction
- personalized systems

### **Adaptation**

Similar techniques can be used to:

- predict customer preferences
- recommend books
- personalize suggestions

## **11. Existing Ontologies and Methodology**

No external ontology was directly reused.

The project follows METHONTOLOGY principles:

- specification
- conceptualization
- formalization
- implementation
- maintenance

## **12. Conclusion**

This project successfully developed a semantic ontology for an online bookstore. The ontology organizes bookstore knowledge in a structured and reusable manner. The integration of semantic technologies improves search, recommendation, and reasoning capabilities. Future work may include larger datasets and automatic ontology population using AI techniques.

## **13. References (APA style)**

- W3C. (2023). OWL Web Ontology Language Overview.
- W3C. (2023). SPARQL Query Language.
- GraphDB Documentation.
- Protégé Ontology Editor Guide.
- Research paper on GRU-RNN ontology systems